

Theory of Algebraic Topology 14

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Lecture notes with worked examples in algebraic topology

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1 Problem 8: The Möbius bundle and its sums/products

Exact statement. Let M be the Möbius bundle over S^1 . Show that $M \oplus M$ is a trivial bundle. Is $M \times M$, considered as a bundle over $S^1 \times S^1$, trivial?

Solution. The Möbius bundle

$$M \rightarrow S^1$$

is the nontrivial real line bundle over S^1 . Its transition function changes sign after going once around the circle. Equivalently, its first Stiefel–Whitney class is nonzero:

$$w_1(M) \neq 0$$

in

$$H^1(S^1; \mathbb{Z}_2) \cong \mathbb{Z}_2.$$

Let

$$w_1(M) = \alpha,$$

where α is the nonzero generator of $H^1(S^1; \mathbb{Z}_2)$.

Now consider the direct sum bundle

$$M \oplus M \rightarrow S^1.$$

The first Stiefel–Whitney class is additive under direct sums:

$$w_1(M \oplus M) = w_1(M) + w_1(M).$$

Therefore

$$w_1(M \oplus M) = \alpha + \alpha = 0,$$

because the coefficients are in $\mathbb{Z}_2$.

Thus $M \oplus M$ is orientable. Since real rank 2 vector bundles over S^1 are classified by whether their monodromy is orientation-preserving or orientation-reversing, orientability implies triviality in this case. Hence

$$M \oplus M \cong S^1 \times \mathbb{R}^2.$$

There is also a concrete transition-function explanation. The Möbius bundle has transition function given by multiplication by -1 . Therefore the transition function of $M \oplus M$ is

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I_2.$$

But $-I_2$ is a rotation by π , so

$$-I_2 \in SO(2).$$

Since $SO(2)$ is connected, $-I_2$ is homotopic to I_2 . Hence the clutching function is null-homotopic, and therefore the bundle is trivial.

Thus

$$\boxed{M \oplus M \cong S^1 \times \mathbb{R}^2.}$$

Now consider

$$M \times M \rightarrow S^1 \times S^1.$$

Here $M \times M$ means the external product bundle. Its fiber over a point

$$(x, y) \in S^1 \times S^1$$

is

$$M_x \times M_y \cong \mathbb{R}^2.$$

Equivalently,

$$M \times M = \pi_1^* M \oplus \pi_2^* M,$$

where

$$\pi_1, \pi_2 : S^1 \times S^1 \rightarrow S^1$$

are the two projection maps.

Using additivity and naturality of the first Stiefel–Whitney class, we get

$$w_1(M \times M) = w_1(\pi_1^* M \oplus \pi_2^* M),$$

so

$$w_1(M \times M) = w_1(\pi_1^* M) + w_1(\pi_2^* M).$$

By naturality,

$$w_1(\pi_1^* M) = \pi_1^* w_1(M),$$

and

$$w_1(\pi_2^* M) = \pi_2^* w_1(M).$$

Therefore

$$w_1(M \times M) = \pi_1^* \alpha + \pi_2^* \alpha.$$

Now

$$H^1(S^1 \times S^1; \mathbb{Z}_2) \cong H^1(S^1; \mathbb{Z}_2) \oplus H^1(S^1; \mathbb{Z}_2) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_2.$$

The class

$$\pi_1^* \alpha + \pi_2^* \alpha$$

is nonzero. Hence

$$w_1(M \times M) \neq 0.$$

Therefore $M \times M$ is not orientable. Since every trivial real vector bundle is orientable, $M \times M$ cannot be trivial.

Thus

$$\boxed{M \times M \rightarrow S^1 \times S^1 \text{ is not trivial.}}$$

2 Problem 9: The unit tangent bundle of S^2

Exact statement. Let

$$E = TS^2$$

be the tangent bundle of S^2 . Show that the unit sphere bundle $S(E)$ is homeomorphic to $SO(3)$, which is also homeomorphic to $\mathbb{RP}^3$. Deduce that

$$e(E) = \pm 2a,$$

where

$$H^2(S^2) = \langle a \rangle.$$

Solution. The unit sphere bundle of TS^2 is the unit tangent bundle

$$S(TS^2) = \{(x, v) \in S^2 \times \mathbb{R}^3 : v \in T_x S^2, \|v\| = 1\}.$$

Since

$$T_x S^2 = \{v \in \mathbb{R}^3 : x \cdot v = 0\},$$

we can write

$$S(TS^2) = \{(x, v) : \|x\| = 1, \|v\| = 1, x \cdot v = 0\}.$$

Thus a point of $S(TS^2)$ is an ordered pair of perpendicular unit vectors:

$$(x, v).$$

From such a pair, we form the ordered triple

$$(x, v, x \times v).$$

This is a positively oriented orthonormal basis of $\mathbb{R}^3$.

Define

$$\Phi : S(TS^2) \rightarrow SO(3)$$

by

$$\Phi(x, v) = \begin{pmatrix} | & | & | \\ x & v & x \times v \\ | & | & | \end{pmatrix}.$$

The columns of this matrix are orthonormal, so the matrix is orthogonal. Moreover, because the basis

$$(x, v, x \times v)$$

is positively oriented, the determinant is 1. Therefore

$$\Phi(x, v) \in SO(3).$$

We now construct the inverse map. Let

$$A \in SO(3).$$

Write the columns of A as

$$A = (u_1, u_2, u_3).$$

Since $A \in SO(3)$, the vectors

$$u_1, u_2, u_3$$

form a positively oriented orthonormal basis of $\mathbb{R}^3$. In particular,

$$u_1 \in S^2.$$

Also,

$$u_1 \cdot u_2 = 0,$$

so

$$u_2 \in T_{u_1}S^2.$$

Since

$$\|u_2\| = 1,$$

we have

$$(u_1, u_2) \in S(TS^2).$$

Thus the inverse map is

$$A \mapsto (u_1, u_2).$$

Therefore

$$\boxed{S(TS^2) \cong SO(3).}$$

Next we recall the standard homeomorphism

$$SO(3) \cong \mathbb{RP}^3.$$

One way to see this is through the double covering

$$S^3 \rightarrow SO(3).$$

The unit quaternions S^3 act on $\mathbb{R}^3$ by rotations, and the two unit quaternions q and $-q$ determine the same rotation. Therefore

$$SO(3) \cong S^3/\{\pm 1\}.$$

But

$$S^3/\{\pm 1\} = \mathbb{RP}^3.$$

Hence

$$\boxed{S(TS^2) \cong SO(3) \cong \mathbb{RP}^3.}$$

Now we deduce the Euler class of $E = TS^2$. Since TS^2 is an oriented real rank 2 vector bundle, its unit sphere bundle

$$S(E) \rightarrow S^2$$

is an oriented circle bundle. Oriented circle bundles over S^2 are classified by their Euler class

$$e(E) \in H^2(S^2; \mathbb{Z}).$$

Since

$$H^2(S^2; \mathbb{Z}) \cong \mathbb{Z},$$

and since

$$H^2(S^2; \mathbb{Z}) = \langle a \rangle,$$

we may write

$$e(E) = ka$$

for some integer k .

We use the Gysin sequence for the oriented circle bundle

$$S(E) \rightarrow S^2.$$

The relevant part of the Gysin sequence is

$$H^0(S^2; \mathbb{Z}) \xrightarrow{\smile e(E)} H^2(S^2; \mathbb{Z}) \longrightarrow H^2(S(E); \mathbb{Z}) \longrightarrow H^1(S^2; \mathbb{Z}).$$

Now

$$H^0(S^2; \mathbb{Z}) \cong \mathbb{Z},$$

$$H^2(S^2; \mathbb{Z}) \cong \mathbb{Z},$$

and

$$H^1(S^2; \mathbb{Z}) = 0.$$

The map

$$H^0(S^2; \mathbb{Z}) \rightarrow H^2(S^2; \mathbb{Z})$$

is multiplication by the Euler class $e(E) = ka$. Therefore the cokernel is

$$H^2(S(E); \mathbb{Z}) \cong \mathbb{Z}/k\mathbb{Z}.$$

But we have shown that

$$S(E) \cong \mathbb{RP}^3.$$

The integral cohomology of $\mathbb{RP}^3$ gives

$$H^2(\mathbb{RP}^3; \mathbb{Z}) \cong \mathbb{Z}_2.$$

Hence

$$H^2(S(E); \mathbb{Z}) \cong \mathbb{Z}_2.$$

Therefore

$$\mathbb{Z}/k\mathbb{Z} \cong \mathbb{Z}_2.$$

It follows that

$$|k| = 2.$$

Thus

$$k = \pm 2.$$

Therefore

$$\boxed{e(E) = \pm 2a.}$$

Since $E = TS^2$, we also write

$$\boxed{e(TS^2) = \pm 2a.}$$

The sign depends on the chosen orientation convention for the generator a .

3 Problem 10: Orthogonal complements of subbundles

Exact statement. Let

$$E \rightarrow B$$

be a real vector bundle equipped with a Riemannian metric, and let

$$F \subset E$$

be a subbundle. Show that $F^\perp$ is a vector bundle, and that

$$F \oplus F^\perp \cong E.$$

Solution. Since $E \rightarrow B$ is equipped with a Riemannian metric, every fiber E_b is a finite-dimensional real inner product space. Since $F \subset E$ is a subbundle, for each $b \in B$, the fiber

$$F_b$$

is a linear subspace of

$$E_b.$$

Define the orthogonal complement fiberwise by

$$F_b^\perp = \{v \in E_b : \langle v, w \rangle_b = 0 \text{ for all } w \in F_b\}.$$

Then define

$$F^\perp = \coprod_{b \in B} F_b^\perp.$$

We want to show that

$$F^\perp \rightarrow B$$

is itself a vector bundle.

Let $U \subset B$ be an open set over which F is trivial. Choose a local frame for F over U :

$$s_1, \dots, s_k.$$

Since $F \subset E$ is a subbundle, we may extend this to a local frame for E over U :

$$s_1, \dots, s_k, s_{k+1}, \dots, s_n.$$

Now apply the Gram–Schmidt process fiber by fiber, using the Riemannian metric on E . Because the metric varies continuously, respectively smoothly, over B , the resulting orthonormal frame also varies continuously, respectively smoothly.

Thus we obtain a local orthonormal frame

$$e_1, \dots, e_n$$

for $E|_U$, such that

$$e_1, \dots, e_k$$

span $F|_U$. Since the frame is orthonormal, the remaining vectors

$$e_{k+1}, \dots, e_n$$

span the orthogonal complement $F^\perp|_U$.

Therefore

$$F^\perp|_U \cong U \times \mathbb{R}^{n-k}.$$

This proves that $F^\perp \rightarrow B$ is locally trivial, and hence $F^\perp$ is a vector bundle.

Finally, for every point $b \in B$, ordinary linear algebra gives the orthogonal decomposition

$$E_b = F_b \oplus F_b^\perp.$$

Thus we define a bundle map

$$\Psi : F \oplus F^\perp \rightarrow E$$

by

$$\Psi(u, v) = u + v.$$

On each fiber,

$$\Psi_b : F_b \oplus F_b^\perp \rightarrow E_b$$

is a linear isomorphism. Since this map is expressed locally in continuous, respectively smooth, frames, it is a vector bundle isomorphism.

Hence

$$\boxed{F^\perp \text{ is a vector bundle}}$$

and

$$\boxed{F \oplus F^\perp \cong E.}$$

Final conclusions

For Problem 8,

$$\boxed{M \oplus M \text{ is trivial, but } M \times M \rightarrow S^1 \times S^1 \text{ is not trivial.}}$$

For Problem 9,

$$\boxed{S(TS^2) \cong SO(3) \cong \mathbb{RP}^3, \quad e(TS^2) = \pm 2a.}$$

For Problem 10,

$$\boxed{F^\perp \text{ is a vector bundle and } F \oplus F^\perp \cong E.}$$

4 Theory for Problems 8–10

The common theme behind Problems 8–10 is that vector bundles are locally trivial objects whose global twisting can be detected by transition functions and characteristic classes. In particular, the Möbius bundle illustrates nontrivial twisting, the tangent bundle of S^2

illustrates the Euler class, and orthogonal complements show how a Riemannian metric allows one to split vector bundles.

Informally, the guiding principle is

vector bundles are locally products, but globally they may be twisted.

The twisting is often measured by characteristic classes such as the Stiefel–Whitney classes and the Euler class.

4.1 Real vector bundles

A real vector bundle of rank n over a topological space B is a continuous map

$$\pi : E \rightarrow B$$

such that every point $b \in B$ has an open neighbourhood $U \subset B$ for which there is a homeomorphism

$$\pi^{-1}(U) \cong U \times \mathbb{R}^n.$$

This homeomorphism must be compatible with the projection to U . In other words, locally the bundle looks like the product

$$U \times \mathbb{R}^n \rightarrow U.$$

The space E is called the total space, B is called the base space, and the fiber over a point $b \in B$ is

$$E_b = \pi^{-1}(b) \cong \mathbb{R}^n.$$

The important point is that a vector bundle is locally trivial but may be globally nontrivial. This means that although every small part of the bundle looks like a product, the whole bundle may be twisted.

The main examples appearing in Problems 8–10 are:

$$M \rightarrow S^1,$$

where M is the Möbius line bundle,

$$TS^2 \rightarrow S^2,$$

where TS^2 is the tangent bundle of the sphere, and

$$F \subset E,$$

where F is a subbundle of a vector bundle $E \rightarrow B$.

4.2 The Möbius bundle

The Möbius bundle

$$M \rightarrow S^1$$

is the simplest example of a nontrivial real line bundle.

A real line bundle has one-dimensional fibers:

$$M_x \cong \mathbb{R}.$$

Locally, the Möbius bundle looks like

$$U \times \mathbb{R}.$$

Globally, however, after travelling once around the base circle S^1 , the fiber coordinate changes sign:

$$v \mapsto -v.$$

This sign reversal is the essential twist of the Möbius bundle.

The trivial real line bundle over S^1 is

$$S^1 \times \mathbb{R} \rightarrow S^1.$$

The Möbius bundle is not isomorphic to this trivial line bundle because it has a global sign reversal after one loop around the circle.

Thus the Möbius bundle is a model example of the difference between local triviality and global triviality.

4.3 Transition functions

A vector bundle can be described using transition functions.

Suppose B is covered by open sets

$$\{U_i\}_{i \in I}$$

such that over each U_i , the bundle $E \rightarrow B$ is trivial:

$$E|_{U_i} \cong U_i \times \mathbb{R}^n.$$

On an overlap

$$U_i \cap U_j,$$

the two trivializations may differ. This difference is encoded by a transition function

$$g_{ij} : U_i \cap U_j \rightarrow GL(n, \mathbb{R}).$$

These maps tell us how to glue the local product pieces together.

For a real line bundle, the structure group is

$$GL(1, \mathbb{R}) = \mathbb{R}^\times.$$

This group has two connected components:

$$\mathbb{R}^\times = \mathbb{R}_{>0} \cup \mathbb{R}_{<0}.$$

The component $\mathbb{R}_{>0}$ preserves orientation of the line, while the component $\mathbb{R}_{<0}$ reverses orientation.

For the Möbius bundle, the essential transition function is represented by

$$-1.$$

That is, after going around the base circle, the fiber coordinate changes by multiplication by -1 . This is why the Möbius bundle is nontrivial.

4.4 Direct sums of vector bundles

If

$$E \rightarrow B$$

and

$$F \rightarrow B$$

are vector bundles over the same base B , then their direct sum is the vector bundle

$$E \oplus F \rightarrow B$$

whose fiber over $b \in B$ is

$$(E \oplus F)_b = E_b \oplus F_b.$$

If E has rank m and F has rank n , then

$$E \oplus F$$

has rank $m + n$.

For example,

$$M \oplus M \rightarrow S^1$$

has fiber

$$(M \oplus M)_x = M_x \oplus M_x \cong \mathbb{R}^2.$$

The key point is that both copies of M twist over the same loop in S^1 .

The transition function of M is

$$-1.$$

Therefore the transition function of $M \oplus M$ is

$$-1 \oplus (-1) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I_2.$$

But $-I_2$ is a rotation by π . Therefore

$$-I_2 \in SO(2).$$

Since $SO(2)$ is connected, there is a continuous path in $SO(2)$ from I_2 to $-I_2$. Explicitly,

$$R(t) = \begin{pmatrix} \cos(\pi t) & -\sin(\pi t) \\ \sin(\pi t) & \cos(\pi t) \end{pmatrix}, \quad 0 \leq t \leq 1,$$

satisfies

$$R(0) = I_2, \quad R(1) = -I_2.$$

Thus the transition function $-I_2$ is homotopic to the identity in the orientation-preserving group $SO(2)$. Consequently, the rank 2 bundle

$$M \oplus M$$

is trivial:

$$M \oplus M \cong S^1 \times \mathbb{R}^2.$$

Geometrically, one can say:

one sign flip gives a Möbius twist, but two sign flips together give a rotation.

4.5 Stiefel–Whitney classes

The Stiefel–Whitney classes are characteristic classes of real vector bundles. For a real vector bundle

$$E \rightarrow B,$$

the i -th Stiefel–Whitney class is

$$w_i(E) \in H^i(B; \mathbb{Z}/2\mathbb{Z}).$$

The most important one for Problems 8–10 is the first Stiefel–Whitney class

$$w_1(E) \in H^1(B; \mathbb{Z}/2\mathbb{Z}).$$

It detects orientability:

$$E \text{ is orientable} \iff w_1(E) = 0.$$

For the Möbius bundle,

$$w_1(M) \neq 0.$$

Indeed,

$$H^1(S^1; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}.$$

Let

$$\alpha \in H^1(S^1; \mathbb{Z}/2\mathbb{Z})$$

be the nonzero element. Then

$$w_1(M) = \alpha.$$

The first Stiefel–Whitney class is additive under direct sums:

$$w_1(E \oplus F) = w_1(E) + w_1(F).$$

Therefore

$$w_1(M \oplus M) = w_1(M) + w_1(M).$$

Since

$$w_1(M) = \alpha,$$

we get

$$w_1(M \oplus M) = \alpha + \alpha.$$

But cohomology is taken with $\mathbb{Z}/2\mathbb{Z}$ -coefficients, so

$$\alpha + \alpha = 0.$$

Hence

$$w_1(M \oplus M) = 0.$$

Thus $M \oplus M$ is orientable. Since it is a rank 2 bundle over S^1 , this orientability is enough to imply that it is trivial.

Therefore

$$\boxed{M \oplus M \cong S^1 \times \mathbb{R}^2.}$$

4.6 External products of vector bundles

The expression

$$M \times M$$

in Problem 8 should not be confused with

$$M \oplus M.$$

The direct sum

$$M \oplus M$$

is a bundle over the same base S^1 . Its fiber over $x \in S^1$ is

$$M_x \oplus M_x.$$

By contrast, the product

$$M \times M$$

is naturally a bundle over

$$S^1 \times S^1.$$

The fiber over $(x, y) \in S^1 \times S^1$ is

$$(M \times M)_{(x,y)} = M_x \times M_y \cong \mathbb{R}^2.$$

This bundle may be written as

$$M \times M = \pi_1^* M \oplus \pi_2^* M,$$

where

$$\pi_1 : S^1 \times S^1 \rightarrow S^1$$

and

$$\pi_2 : S^1 \times S^1 \rightarrow S^1$$

are the projection maps.

Here $\pi_1^* M$ twists in the first S^1 -direction, while $\pi_2^* M$ twists in the second S^1 -direction.

This is the key difference:

$$M \oplus M$$

has two twists over the same circle, so the first Stiefel–Whitney classes cancel.

But

$$M \times M \rightarrow S^1 \times S^1$$

has one twist in each independent direction of the torus. These twists do not cancel because they live in different cohomology directions.

Indeed,

$$H^1(S^1 \times S^1; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}.$$

If

$$\alpha \in H^1(S^1; \mathbb{Z}/2\mathbb{Z})$$

is the nonzero class, then

$$\pi_1^* \alpha$$

and

$$\pi_2^* \alpha$$

are the two independent generators of

$$H^1(S^1 \times S^1; \mathbb{Z}/2\mathbb{Z}).$$

Therefore

$$w_1(M \times M) = \pi_1^* \alpha + \pi_2^* \alpha.$$

This class is nonzero. Hence

$$M \times M$$

is not orientable. Since every trivial real vector bundle is orientable, it follows that

$$M \times M \rightarrow S^1 \times S^1$$

is not trivial.

Thus

$M \times M \rightarrow S^1 \times S^1 \text{ is not trivial.}$

4.7 Tangent bundles

Let X be a smooth manifold. The tangent bundle of X is

$$TX \rightarrow X.$$

The fiber over a point $x \in X$ is the tangent space

$$T_x X.$$

Thus

$$TX = \coprod_{x \in X} T_x X.$$

For the sphere

$$S^2 \subset \mathbb{R}^3,$$

the tangent space at $x \in S^2$ is

$$T_x S^2 = \{v \in \mathbb{R}^3 : v \cdot x = 0\}.$$

This is the plane through the origin perpendicular to x .

Therefore the tangent bundle of S^2 can be written as

$$TS^2 = \{(x, v) \in S^2 \times \mathbb{R}^3 : x \cdot v = 0\}.$$

The projection map is

$$TS^2 \rightarrow S^2, \quad (x, v) \mapsto x.$$

Since each fiber $T_x S^2$ is a two-dimensional real vector space, the tangent bundle

$$TS^2 \rightarrow S^2$$

is a real rank 2 vector bundle.

4.8 Unit sphere bundles

Let

$$E \rightarrow B$$

be a real vector bundle equipped with a metric. The unit sphere bundle of E is

$$S(E) = \{v \in E : \|v\| = 1\}.$$

If E has rank n , then each fiber of $S(E)$ is

$$S^{n-1}.$$

Indeed, for every $b \in B$,

$$S(E)_b = \{v \in E_b : \|v\| = 1\} \cong S^{n-1}.$$

For $E = TS^2$, the tangent bundle has rank 2. Hence

$$S(TS^2) \rightarrow S^2$$

is a bundle with fiber

$$S^1.$$

This is the unit tangent bundle of S^2 .

A point of $S(TS^2)$ is a pair

$$(x, v),$$

where

$$x \in S^2$$

and

$$v \in T_x S^2$$

is a unit tangent vector.

Equivalently,

$$\|x\| = 1, \quad \|v\| = 1, \quad x \cdot v = 0.$$

Thus

$$S(TS^2) = \{(x, v) \in S^2 \times \mathbb{R}^3 : \|v\| = 1, x \cdot v = 0\}.$$

4.9 Why $S(TS^2) \cong SO(3)$

The group

$$SO(3)$$

is the group of orientation-preserving orthogonal transformations of $\mathbb{R}^3$. Equivalently, an element of $SO(3)$ is an ordered positively oriented orthonormal basis of $\mathbb{R}^3$.

A point of $S(TS^2)$ is a pair

$$(x, v),$$

where x and v are perpendicular unit vectors in $\mathbb{R}^3$. From these two vectors, we can construct a third vector

$$x \times v.$$

The triple

$$(x, v, x \times v)$$

is a positively oriented orthonormal basis of $\mathbb{R}^3$. Hence it determines an element of $SO(3)$.

Therefore define

$$\Phi : S(TS^2) \rightarrow SO(3)$$

by

$$\Phi(x, v) = \begin{pmatrix} | & | & | \\ x & v & x \times v \\ | & | & | \end{pmatrix}.$$

The columns are orthonormal, so the matrix is orthogonal. Moreover, because the ordered basis

$$(x, v, x \times v)$$

is positively oriented, the determinant is 1. Therefore

$$\Phi(x, v) \in SO(3).$$

Conversely, let

$$A \in SO(3).$$

Write the columns of A as

$$A = (u_1, u_2, u_3).$$

Since $A \in SO(3)$, the vectors u_1, u_2, u_3 form a positively oriented orthonormal basis of $\mathbb{R}^3$. In particular,

$$u_1 \in S^2,$$

and

$$u_1 \cdot u_2 = 0.$$

Therefore

$$u_2 \in T_{u_1}S^2.$$

Since

$$\|u_2\| = 1,$$

we get

$$(u_1, u_2) \in S(TS^2).$$

Thus the inverse map is

$$A \mapsto (u_1, u_2).$$

Therefore

$$S(TS^2) \cong SO(3).$$

Geometrically, this means

the unit tangent bundle of S^2 is the space of positively oriented orthonormal frames in $\mathbb{R}^3$.

4.10 Why $SO(3) \cong \mathbb{RP}^3$

The group $SO(3)$ is homeomorphic to real projective 3-space:

$$SO(3) \cong \mathbb{RP}^3.$$

One way to understand this is through unit quaternions.

The unit quaternions form a space homeomorphic to S^3 . Every unit quaternion determines a rotation of $\mathbb{R}^3$. However, the two opposite unit quaternions

$$q$$

and

$$-q$$

determine the same rotation.

Therefore there is a double covering

$$S^3 \rightarrow SO(3),$$

where the fibers consist of pairs

$$\{q, -q\}.$$

Hence

$$SO(3) \cong S^3 / \{q \sim -q\}.$$

But by definition,

$$S^3 / \{q \sim -q\} = \mathbb{RP}^3.$$

Therefore

$$\boxed{SO(3) \cong \mathbb{RP}^3.}$$

Combining this with the previous result gives

$$\boxed{S(TS^2) \cong SO(3) \cong \mathbb{RP}^3.}$$

4.11 The Euler class

The Euler class is a characteristic class of oriented real vector bundles.

If

$$E \rightarrow B$$

is an oriented real vector bundle of rank n , then the Euler class is

$$e(E) \in H^n(B; \mathbb{Z}).$$

The Euler class measures the obstruction to finding a nowhere-zero section of E .

A section of $E \rightarrow B$ is a continuous or smooth map

$$s : B \rightarrow E$$

such that

$$\pi \circ s = \text{id}_B.$$

Equivalently, for each $b \in B$, the section chooses a vector

$$s(b) \in E_b.$$

A nowhere-zero section is a section such that

$$s(b) \neq 0$$

for every $b \in B$.

If E has a nowhere-zero section, then

$$e(E) = 0.$$

Thus a nonzero Euler class prevents the existence of a nowhere-zero section.

For the tangent bundle of a closed oriented n -manifold X , the Euler class is related to the Euler characteristic by the formula

$$\langle e(TX), [X] \rangle = \chi(X).$$

Here

$$[X] \in H_n(X; \mathbb{Z})$$

is the fundamental class of X , and

$$\chi(X)$$

is the Euler characteristic.

For $X = S^2$, we have

$$\chi(S^2) = 2.$$

Therefore

$$\langle e(TS^2), [S^2] \rangle = 2.$$

Since

$$H^2(S^2; \mathbb{Z}) \cong \mathbb{Z},$$

if

$$H^2(S^2; \mathbb{Z}) = \langle a \rangle,$$

then

$$e(TS^2) = \pm 2a.$$

The sign depends on whether a agrees with the chosen orientation of S^2 .

Thus

$$e(TS^2) = \pm 2a.$$

This is also the cohomological form of the hairy ball theorem:

$$\text{there is no nowhere-zero continuous tangent vector field on } S^2.$$

4.12 The Euler class from the Gysin sequence

Problem 9 asks us to deduce

$$e(TS^2) = \pm 2a$$

from the fact that

$$S(TS^2) \cong \mathbb{RP}^3.$$

This can be done using the Gysin sequence.

Let

$$S^1 \rightarrow S(E) \rightarrow B$$

be an oriented circle bundle. Then there is a long exact cohomology sequence called the Gysin sequence. The part relevant to Problem 9 is

$$H^0(B; \mathbb{Z}) \xrightarrow{\smile e(E)} H^2(B; \mathbb{Z}) \rightarrow H^2(S(E); \mathbb{Z}) \rightarrow H^1(B; \mathbb{Z}).$$

For the unit tangent bundle

$$S(TS^2) \rightarrow S^2,$$

this becomes

$$H^0(S^2; \mathbb{Z}) \xrightarrow{\smile e(TS^2)} H^2(S^2; \mathbb{Z}) \rightarrow H^2(S(TS^2); \mathbb{Z}) \rightarrow H^1(S^2; \mathbb{Z}).$$

Now

$$H^0(S^2; \mathbb{Z}) \cong \mathbb{Z},$$

$$H^2(S^2; \mathbb{Z}) \cong \mathbb{Z},$$

and

$$H^1(S^2; \mathbb{Z}) = 0.$$

Let

$$e(TS^2) = ka,$$

where

$$H^2(S^2; \mathbb{Z}) = \langle a \rangle.$$

Then the map

$$H^0(S^2; \mathbb{Z}) \rightarrow H^2(S^2; \mathbb{Z})$$

is multiplication by k . Therefore its cokernel is

$$\mathbb{Z}/k\mathbb{Z}.$$

Exactness of the Gysin sequence gives

$$H^2(S(TS^2); \mathbb{Z}) \cong \mathbb{Z}/k\mathbb{Z}.$$

But we know that

$$S(TS^2) \cong \mathbb{RP}^3.$$

The integral cohomology of $\mathbb{RP}^3$ contains

$$H^2(\mathbb{RP}^3; \mathbb{Z}) \cong \mathbb{Z}_2.$$

Therefore

$$\mathbb{Z}/k\mathbb{Z} \cong \mathbb{Z}_2.$$

Hence

$$|k| = 2.$$

So

$$k = \pm 2.$$

Therefore

$$\boxed{e(TS^2) = \pm 2a.}$$

This is exactly the result requested in Problem 9.

4.13 Riemannian metrics on vector bundles

A Riemannian metric on a real vector bundle

$$E \rightarrow B$$

is a smoothly or continuously varying inner product on each fiber:

$$\langle -, - \rangle_b : E_b \times E_b \rightarrow \mathbb{R}.$$

Thus for every point $b \in B$, the fiber E_b becomes an inner product space.

This allows us to speak about lengths:

$$\|v\|_b = \sqrt{\langle v, v \rangle_b},$$

angles, and orthogonality inside each fiber.

In particular, if

$$v, w \in E_b,$$

then we say v is orthogonal to w if

$$\langle v, w \rangle_b = 0.$$

This notion is essential in Problem 10, where we define the orthogonal complement of a subbundle.

4.14 Subbundles

Let

$$E \rightarrow B$$

be a vector bundle. A subbundle

$$F \subset E$$

means that for each point $b \in B$, the fiber

$$F_b$$

is a linear subspace of

$$E_b,$$

and these subspaces vary continuously or smoothly with b .

It is important that F is not merely a collection of subspaces. It must itself be a vector bundle over B .

If

$$\text{rank}(E) = n$$

and

$$\text{rank}(F) = k,$$

then locally we can choose a frame

$$s_1, \dots, s_k$$

for F . Because $F \subset E$ is a subbundle, this frame can be extended locally to a frame

$$s_1, \dots, s_k, s_{k+1}, \dots, s_n$$

for E .

This local extension property is what allows us to construct complements.

4.15 Orthogonal complements

Let

$$E \rightarrow B$$

be a real vector bundle with a Riemannian metric, and let

$$F \subset E$$

be a subbundle.

For each point $b \in B$, define

$$F_b^\perp = \{v \in E_b : \langle v, w \rangle_b = 0 \text{ for all } w \in F_b\}.$$

Then define

$$F^\perp = \coprod_{b \in B} F_b^\perp.$$

Fiberwise, ordinary linear algebra gives

$$E_b = F_b \oplus F_b^\perp.$$

However, Problem 10 asks for more than this fiberwise statement. It asks us to prove that

$$F^\perp$$

is itself a vector bundle over B .

To see this, let $U \subset B$ be an open set over which F is trivial. Choose a local frame

$$s_1, \dots, s_k$$

for $F|_U$. Since $F \subset E$ is a subbundle, extend this to a local frame

$$s_1, \dots, s_k, s_{k+1}, \dots, s_n$$

for $E|_U$.

Now apply the Gram–Schmidt process fiber by fiber using the Riemannian metric. We obtain an orthonormal frame

$$e_1, \dots, e_n$$

for $E|_U$, such that

$$e_1, \dots, e_k$$

span $F|_U$.

Because the frame is orthonormal, the remaining vectors

$$e_{k+1}, \dots, e_n$$

span

$$F^\perp|_U.$$

Therefore

$$F^\perp|_U \cong U \times \mathbb{R}^{n-k}.$$

This proves that $F^\perp \rightarrow B$ is locally trivial, hence $F^\perp$ is a vector bundle.

Furthermore, define

$$\Psi : F \oplus F^\perp \rightarrow E$$

by

$$\Psi(u, v) = u + v.$$

For each $b \in B$, the fiber map is

$$\Psi_b : F_b \oplus F_b^\perp \rightarrow E_b, \quad (u, v) \mapsto u + v.$$

This is a linear isomorphism by the orthogonal decomposition

$$E_b = F_b \oplus F_b^\perp.$$

Since this construction is local and compatible with local frames, Ψ is an isomorphism of vector bundles.

Therefore

$$\boxed{F^\perp \text{ is a vector bundle}}$$

and

$$\boxed{F \oplus F^\perp \cong E.}$$

4.16 The splitting principle behind Problem 10

Problem 10 is an example of a general principle:

$$\boxed{\text{short exact sequences of vector bundles split when a metric is available.}}$$

Suppose we have a short exact sequence of vector bundles

$$0 \rightarrow F \rightarrow E \rightarrow Q \rightarrow 0.$$

This means that F is a subbundle of E , and Q is the quotient bundle

$$Q = E/F.$$

If E has a Riemannian metric, then the quotient bundle E/F can be identified with the orthogonal complement $F^\perp$.

Indeed, every vector $e \in E_b$ decomposes uniquely as

$$e = f + v,$$

where

$$f \in F_b, \quad v \in F_b^\perp.$$

Thus the quotient class of e modulo F_b is represented uniquely by its orthogonal component $v \in F_b^\perp$. This gives a natural isomorphism

$$E/F \cong F^\perp.$$

Hence

$$E \cong F \oplus F^\perp.$$

This is why Riemannian metrics are so useful in vector bundle theory: they allow one to split bundles into orthogonal pieces.

However, one should remember that the orthogonal complement depends on the chosen metric. Different metrics may give different subbundles $F^\perp$, although all such complements are isomorphic to the quotient bundle E/F .

4.17 How Problems 8–10 are connected

The three problems are connected by the same general philosophy: vector bundles are locally simple, but their global behaviour is controlled by twisting.

Problem 8 studies how twisting behaves under direct sums and products.

The bundle

$$M \oplus M$$

is trivial because the two copies of the Möbius twist occur over the same circle and cancel in $\mathbb{Z}/2\mathbb{Z}$ -cohomology:

$$w_1(M \oplus M) = w_1(M) + w_1(M) = 0.$$

However,

$$M \times M \rightarrow S^1 \times S^1$$

is not trivial because the two twists occur in two independent directions of the torus:

$$w_1(M \times M) = \pi_1^* \alpha + \pi_2^* \alpha \neq 0.$$

Problem 9 studies a geometric vector bundle:

$$TS^2 \rightarrow S^2.$$

Its unit sphere bundle is the space of unit tangent vectors on S^2 . This space is naturally identified with

$$SO(3),$$

and hence with

$$\mathbb{RP}^3.$$

The Gysin sequence then forces the Euler class to be

$$e(TS^2) = \pm 2a.$$

Problem 10 studies how a Riemannian metric gives a decomposition of a bundle into a subbundle and its orthogonal complement:

$$E \cong F \oplus F^\perp.$$

Together, the three problems illustrate the following principles:

transition functions describe how vector bundles are glued.

Stiefel–Whitney classes detect real bundle twisting and orientability.

Euler classes detect obstructions to nowhere-zero sections.

Riemannian metrics allow subbundles to have orthogonal complements.

Thus, the overall lesson is:

metrics help split vector bundles, while characteristic classes detect when bundles cannot be globally

5 Examples of sections: step-by-step

1) Trivial bundle $E = B \times \mathbb{R}^n$

The projection is

$$\pi : B \times \mathbb{R}^n \rightarrow B, \quad \pi(b, v) = b.$$

A section is any map $s : B \rightarrow B \times \mathbb{R}^n$ of the form

$$s(b) = (b, f(b))$$

for some function $f : B \rightarrow \mathbb{R}^n$. Then

$$(\pi \circ s)(b) = \pi(b, f(b)) = b,$$

so $\pi \circ s = \text{id}_B$.

A *nowhere-zero* section is obtained by choosing a constant nonzero vector $v_0 \neq 0$ and setting

$$s(b) = (b, v_0).$$

2) Tangent bundle of $\mathbb{R}^n$

Using the standard identification

$$T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n,$$

a section is the same as a vector field $X : \mathbb{R}^n \rightarrow \mathbb{R}^n$, namely

$$s(x) = (x, X(x)).$$

Then $\pi(x, X(x)) = x$, so $\pi \circ s = \text{id}_{\mathbb{R}^n}$.

A nowhere-zero example is the constant vector field $X(x) = e_1$.

3) Tangent bundle of S^1 : a nowhere-zero section exists

Parametrize $S^1 \subset \mathbb{R}^2$ by $(\cos \theta, \sin \theta)$. A standard tangent vector field is

$$X(\cos \theta, \sin \theta) = (-\sin \theta, \cos \theta),$$

which is tangent and never zero. Hence it gives a nowhere-zero section of TS^1 .

4) Tangent bundle of S^2 : no nowhere-zero section

There is no continuous nowhere-zero section of TS^2 (the hairy ball theorem). Thus one can have sections locally (in charts), but not a global nowhere-zero one.

This is a typical kind of obstruction detected by characteristic classes such as the Stiefel–Whitney classes (and, in oriented settings, the Euler class).

5) Möbius line bundle over S^1 : nonzero section exists, but no global trivialization

Think of the Möbius bundle as a real line over each point of S^1 that flips sign after one loop.

It *does* admit a nowhere-zero section (for instance, choose a unit vector in each fiber continuously). However, it is *not* isomorphic to the trivial bundle $S^1 \times \mathbb{R}$: it is a nontrivial line bundle.

So:

$$\text{“nowhere-zero section”} \not\Rightarrow \text{“bundle trivial.”}$$

6) Covering space example (discrete fiber): no global section

Let

$$\pi : \mathbb{R} \rightarrow S^1, \quad \pi(t) = e^{2\pi it}.$$

A section would be a continuous map $s : S^1 \rightarrow \mathbb{R}$ such that $\pi \circ s = \text{id}_{S^1}$. Such a global section cannot exist, because it would continuously choose a logarithm/argument around the circle.

This is a clean example where the obstruction is *monodromy*.

5.1 Sections of bundles: definition and basic checks

Let $\pi : E \rightarrow B$ be a (vector) bundle. A *section* is a map

$$s : B \rightarrow E$$

such that

$$\pi \circ s = \text{id}_B.$$

Equivalently, for each $b \in B$ the section chooses a point in the fiber:

$$s(b) \in E_b := \pi^{-1}(b).$$

If $E \rightarrow B$ is a *vector* bundle, we call s *nowhere-zero* if

$$s(b) \neq 0 \in E_b \quad \text{for all } b \in B.$$

5.2 Step-by-step examples of sections (and obstructions)

1. Trivial bundle $E = B \times \mathbb{R}^n$

Step 1 (projection). Define

$$\pi : B \times \mathbb{R}^n \rightarrow B, \quad \pi(b, v) = b.$$

Step 2 (define a section). Given any map $f : B \rightarrow \mathbb{R}^n$, set

$$s(b) = (b, f(b)).$$

Step 3 (verify $\pi \circ s = \text{id}_B$). For each $b \in B$,

$$(\pi \circ s)(b) = \pi(b, f(b)) = b,$$

so indeed $\pi \circ s = \text{id}_B$.

Step 4 (nowhere-zero). The section s is nowhere-zero iff $f(b) \neq 0$ for all b . For example, choosing a fixed $v_0 \in \mathbb{R}^n$ with $v_0 \neq 0$ gives

$$s(b) = (b, v_0),$$

which is nowhere-zero.

2. Tangent bundle of $\mathbb{R}^n$

Step 1 (identify $T\mathbb{R}^n$). There is a standard identification

$$T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n,$$

with projection $\pi(x, v) = x$.

Step 2 (define a section). A section is exactly a vector field $X : \mathbb{R}^n \rightarrow \mathbb{R}^n$:

$$s(x) = (x, X(x)).$$

Step 3 (verify). For each $x \in \mathbb{R}^n$,

$$(\pi \circ s)(x) = \pi(x, X(x)) = x,$$

so $\pi \circ s = \text{id}_{\mathbb{R}^n}$.

Step 4 (nowhere-zero). If $X(x) \neq 0$ for all x , then s is nowhere-zero. Example: $X(x) = e_1$ (a constant nonzero vector field).

3. Tangent bundle of S^1 (a nowhere-zero section exists)

Write $S^1 = \{(\cos \theta, \sin \theta) : \theta \in \mathbb{R}\} \subset \mathbb{R}^2$.

Step 1 (bundle + projection). The tangent bundle $\pi : TS^1 \rightarrow S^1$ sends a tangent vector to its base point.

Step 2 (define a section / vector field). Define a tangent vector field by

$$X(\cos \theta, \sin \theta) = (-\sin \theta, \cos \theta).$$

This determines a section $s : S^1 \rightarrow TS^1$ by $s(p) = (p, X(p))$.

Step 3 (verify). For each $p \in S^1$,

$$\pi(s(p)) = \pi(p, X(p)) = p,$$

so $\pi \circ s = \text{id}_{S^1}$.

Step 4 (nowhere-zero). One has

$$\|X(\cos \theta, \sin \theta)\| = \sqrt{\sin^2 \theta + \cos^2 \theta} = 1 \neq 0,$$

so this section is nowhere-zero.

4. Tangent bundle of S^2 (no nowhere-zero section)

Step 1. A section $s : S^2 \rightarrow TS^2$ is the same as a continuous tangent vector field on S^2 .

Step 2. “Nowhere-zero” means a vector field X with $X(p) \neq 0$ for every $p \in S^2$.

Step 3 (obstruction). The *hairy ball theorem* states that every continuous tangent vector field on S^2 must vanish somewhere.

Conclusion. Hence TS^2 admits no global nowhere-zero section.

5. The Möbius line bundle over S^1 (nowhere-zero section exists, but the bundle is nontrivial)

Let $\pi : E \rightarrow S^1$ be the Möbius line bundle, whose fiber E_p over each $p \in S^1$ is a 1-dimensional real vector space, but with a twist around the circle.

Step 1 (what a section is). A section chooses a vector in each fiber:

$$s(p) \in E_p, \quad p \in S^1,$$

and automatically satisfies $\pi(s(p)) = p$.

Step 2 (nowhere-zero). One can choose a continuous *unit* vector in each fiber (so $s(p) \neq 0$ for all p). Thus a nowhere-zero section exists.

Important point. Existence of a nowhere-zero section does *not* imply the bundle is trivial: the Möbius bundle is not isomorphic to $S^1 \times \mathbb{R}$, even though it has a nowhere-zero section.

6. A though it has a nowhere-zero section.

6. A covering map with no global section: $\pi : \mathbb{R} \rightarrow S^1$

Consider the covering map

$$\pi : \mathbb{R} \rightarrow S^1, \quad \pi(t) = e^{2\pi it}.$$

Step 1 (what a section would mean). A section would be a continuous map $s : S^1 \rightarrow \mathbb{R}$ such that

$$\pi(s(z)) = z \quad \text{for all } z \in S^1.$$

Equivalently, s would continuously choose a real “argument” (a lift) of each point of the circle.

Step 2 (why impossible). Going once around S^1 changes any lift by an integer (by $+1$). A continuous single-valued choice cannot return to the same value after one loop.

Conclusion. Therefore no global section s exists for $\pi : \mathbb{R} \rightarrow S^1$.

Summary (what these examples illustrate)

- In a trivial bundle $B \times \mathbb{R}^n \rightarrow B$, sections are exactly maps $f : B \rightarrow \mathbb{R}^n$.

- For tangent bundles, sections are vector fields; on S^1 there is a nowhere-zero field, but on S^2 not.
- A nowhere-zero section does not necessarily trivialize the bundle (Möbius line bundle).
- Some bundle-like maps (e.g. coverings) have no global section for topological reasons (monodromy).

6 Charts and Diagrams for Problems 8–10

The following diagrams summarize the geometry and topology behind Problems 8–10. The main ideas are:

Möbius twisting, unit tangent bundles, Euler classes, and orthogonal splittings.

6.1 Chart: M , $M \oplus M$, and $M \times M$

The Möbius bundle $M \rightarrow S^1$, its direct sum $M \oplus M$, and its external product $M \times M$ behave differently.

Bundle	Base	Fiber	Trivial?
M	S^1	$\mathbb{R}$	No
$M \oplus M$	S^1	$\mathbb{R}^2$	Yes
$M \times M$	$S^1 \times S^1$	$\mathbb{R}^2$	No

The reason is that

$$w_1(M) = \alpha \neq 0,$$

but

$$w_1(M \oplus M) = \alpha + \alpha = 0.$$

However,

$$w_1(M \times M) = \pi_1^* \alpha + \pi_2^* \alpha \neq 0.$$

Thus the two twists cancel in $M \oplus M$, but they do not cancel in $M \times M$.

6.2 Diagram: One Möbius twist versus two sign reversals

For the Möbius bundle M , going once around S^1 changes the fiber coordinate by

$$v \longmapsto -v.$$

For $M \oplus M$, going once around S^1 changes

$$(v_1, v_2) \longmapsto (-v_1, -v_2).$$

That is the linear transformation

$$-I_2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Therefore

$$\begin{aligned} M & : v \longmapsto -v \\ M \oplus M & : (v_1, v_2) \longmapsto (-v_1, -v_2) \\ & \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \longmapsto \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}. \end{aligned}$$

But

$$-I_2 = R_\pi,$$

where

$$R_\pi = \begin{pmatrix} \cos \pi & -\sin \pi \\ \sin \pi & \cos \pi \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $-I_2$ is a rotation by π . Therefore

$$-I_2 \in SO(2).$$

The geometric slogan is:

two sign reversals become one rotation.

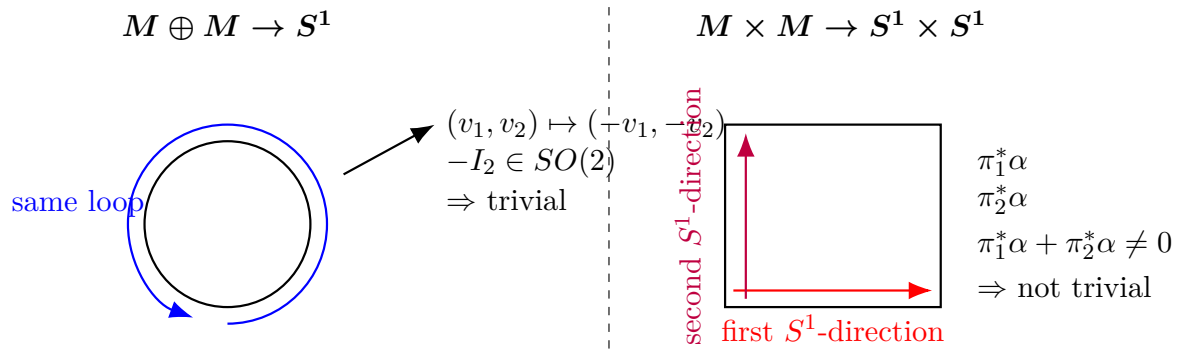
6.3 Diagram: Direct sum versus product bundle

The following diagram shows the difference between

$$M \oplus M \rightarrow S^1$$

and

$$M \times M \rightarrow S^1 \times S^1.$$



Conceptually,

$M \oplus M$: two twists over the same loop cancel,

whereas

$M \times M$: the twists occur in two independent torus directions.

6.4 Diagram: $S(TS^2) \cong SO(3)$

A point of $S(TS^2)$ is a pair

$$(x, v),$$

where

$$x \in S^2, \quad v \in T_x S^2, \quad \|v\| = 1.$$

Since $v \in T_x S^2$, we have

$$x \cdot v = 0.$$

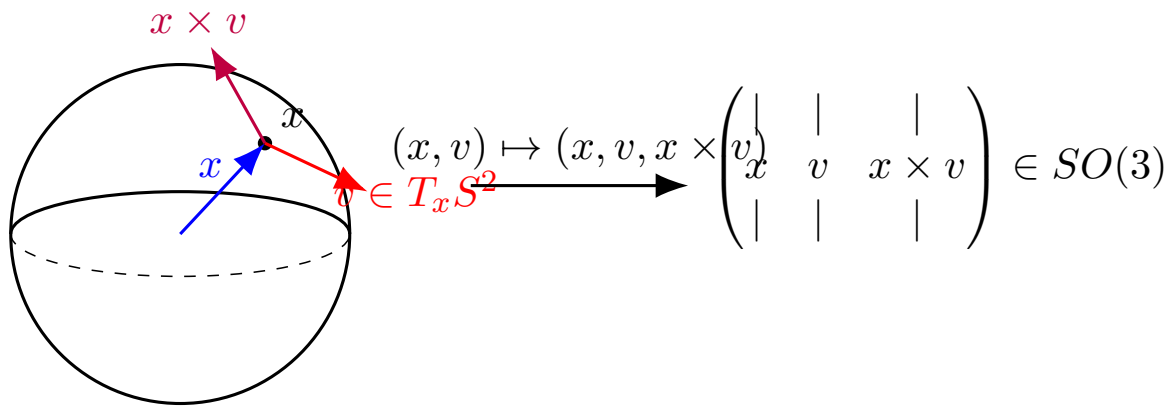
Therefore x and v are perpendicular unit vectors. We can form the third vector

$$x \times v.$$

Thus

$$(x, v) \longmapsto (x, v, x \times v).$$

This gives a positively oriented orthonormal frame in $\mathbb{R}^3$, hence an element of $SO(3)$.



The main meaning is:

$$S(TS^2) = \{\text{unit tangent vectors on } S^2\} \cong \{\text{oriented orthonormal frames in } \mathbb{R}^3\} = SO(3).$$

6.5 Diagram: $SO(3) \cong \mathbb{RP}^3$

The group $SO(3)$ is homeomorphic to real projective 3-space.

The reason is that there is a double covering

$$S^3 \rightarrow SO(3).$$

The two points q and $-q$ in S^3 determine the same rotation. Hence

$$SO(3) \cong S^3 / \{q \sim -q\}.$$

But

$$S^3 / \{q \sim -q\} = \mathbb{RP}^3.$$

Therefore

$$SO(3) \cong \mathbb{RP}^3.$$

$$\begin{array}{ccc} S^3 & \xrightarrow{\text{double cover}} & SO(3) \xleftarrow{\cong} \mathbb{RP}^3 \\ q \sim -q & & S^3 / (q \sim -q) \\ SO(3) \cong S^3 / \{\pm 1\} & & \end{array}$$

Combining the two identifications, we obtain

$$S(TS^2) \cong SO(3) \cong \mathbb{RP}^3.$$

6.6 Flow chart: Euler class from the Gysin sequence

For Problem 9, the deduction of the Euler class is:

$$S(TS^2) \cong \mathbb{RP}^3.$$

Therefore

$$H^2(S(TS^2); \mathbb{Z}) \cong H^2(\mathbb{RP}^3; \mathbb{Z}) \cong \mathbb{Z}_2.$$

On the other hand, the Gysin sequence says that if

$$e(TS^2) = ka,$$

then

$$H^2(S(TS^2); \mathbb{Z}) \cong \mathbb{Z}/k\mathbb{Z}.$$

Thus

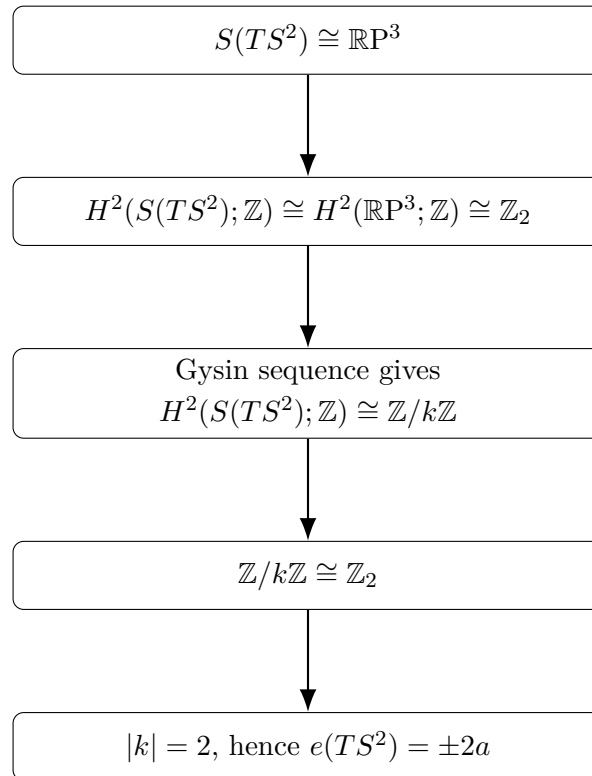
$$\mathbb{Z}/k\mathbb{Z} \cong \mathbb{Z}_2.$$

Hence

$$|k| = 2.$$

Therefore

$$e(TS^2) = \pm 2a.$$



6.7 Diagram: The orthogonal complement $F^\perp$

For Problem 10, the central picture is fiberwise linear algebra.

At each point $b \in B$, we have a fiber E_b . Inside it, the subbundle gives a subspace

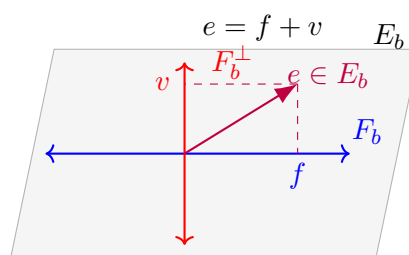
$$F_b \subset E_b.$$

Using the Riemannian metric, we define the orthogonal complement

$$F_b^\perp = \{v \in E_b : \langle v, w \rangle_b = 0 \text{ for all } w \in F_b\}.$$

Then

$$E_b = F_b \oplus F_b^\perp.$$



This diagram shows that every vector

$$e \in E_b$$

decomposes uniquely as

$$e = f + v,$$

where

$$f \in F_b, \quad v \in F_b^\perp.$$

Thus, fiberwise,

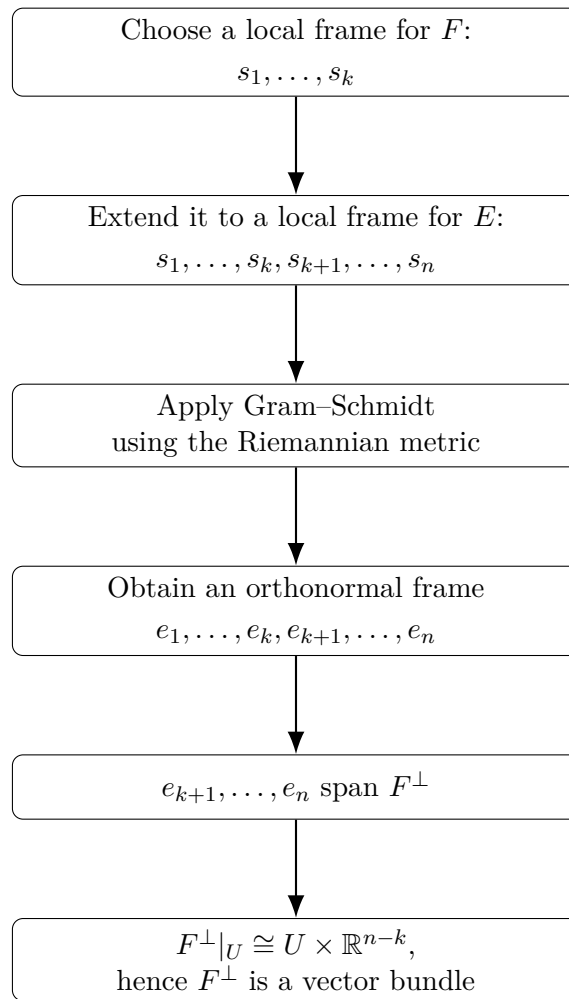
$$E_b = F_b \oplus F_b^\perp.$$

Globally, this becomes the vector bundle isomorphism

$$\boxed{E \cong F \oplus F^\perp.}$$

6.8 Flow chart: Why $F^\perp$ is a vector bundle

The fact that $F_b^\perp$ is a vector space for each $b \in B$ is only a fiberwise statement. To show that $F^\perp \rightarrow B$ is a vector bundle, we use local frames and Gram–Schmidt.



Therefore the orthogonal complement is not only defined fiber by fiber; it varies locally trivially over the base space.

6.9 Summary chart connecting Problems 8–10

Problem	Main object	Main tool	Conclusion
8	$M \rightarrow S^1$	w_1 , transition functions	$M \oplus M$ trivial, but $M \otimes M$ not trivial
9	$TS^2 \rightarrow S^2$	$SO(3)$, $\mathbb{RP}^3$, Gysin sequence	$e(TS^2) = \pm 2a$
10	$F \subset E$	Riemannian metric, Gram-Schmidt	$E \cong F \oplus F^\perp$

6.10 Global conceptual summary

The three problems together show how vector bundles are studied geometrically and cohomologically.

Problem 8: twisting is detected by w_1 .

Problem 9: the Euler class is detected by the topology of the sphere bundle.

Problem 10: a metric gives an orthogonal splitting.

More compactly:

transition functions describe twisting;

characteristic classes detect nontriviality;

Riemannian metrics allow subbundles to split.

Thus the main lesson is

metrics help split vector bundles, while characteristic classes detect when bundles cannot be trivialized

7 Problems 11–12: Clutching construction and sphere bundles over S^1

7.1 Problem 11: Vector bundles over S^n by clutching functions

Problem 11. Given a continuous map

$$\gamma : S^{n-1} \longrightarrow O(k),$$

let

$$E_\gamma = D_N^n \times \mathbb{R}^k \sqcup D_S^n \times \mathbb{R}^k / \sim,$$

where the equivalence relation identifies

$$(x, v) \in S_N^{n-1} \times \mathbb{R}^k$$

with

$$(x, \gamma(x)v) \in S_S^{n-1} \times \mathbb{R}^k.$$

Show that E_γ is a vector bundle over S^n , and that every vector bundle over S^n is isomorphic to some E_γ . Show further that

$$E_\gamma \cong E_{\gamma'} \iff \gamma \simeq \gamma'.$$

Solution.

We view S^n as the union of two closed hemispheres

$$S^n = D_N^n \cup D_S^n,$$

where

$$D_N^n \cap D_S^n = S^{n-1}.$$

The construction glues two trivial vector bundles

$$D_N^n \times \mathbb{R}^k \quad \text{and} \quad D_S^n \times \mathbb{R}^k$$

along their common boundary by the map

$$\gamma : S^{n-1} \rightarrow O(k).$$

Define a projection

$$\pi : E_\gamma \longrightarrow S^n$$

by

$$\pi([x, v]) = x.$$

This is well-defined because the gluing relation identifies only points lying over the same base point $x \in S^{n-1}$. Hence the fiber over $x \in S^n$ is naturally a copy of $\mathbb{R}^k$.

Over the northern hemisphere D_N^n , the bundle is trivial:

$$\pi^{-1}(D_N^n) \cong D_N^n \times \mathbb{R}^k.$$

Similarly, over the southern hemisphere,

$$\pi^{-1}(D_S^n) \cong D_S^n \times \mathbb{R}^k.$$

On the overlap S^{n-1} , the transition function is precisely

$$\gamma(x) \in O(k) \subset GL(k, \mathbb{R}).$$

Since each $\gamma(x)$ is a linear isomorphism of $\mathbb{R}^k$, the vector space structures on the fibers are compatible with the gluing. Therefore $E_\gamma \rightarrow S^n$ is a rank k real vector bundle.

Now let

$$p : E \rightarrow S^n$$

be any rank k real vector bundle. Since both hemispheres D_N^n and D_S^n are contractible, the restrictions

$$E|_{D_N^n} \quad \text{and} \quad E|_{D_S^n}$$

are trivial. Choose trivializations

$$\varphi_N : E|_{D_N^n} \longrightarrow D_N^n \times \mathbb{R}^k$$

and

$$\varphi_S : E|_{D_S^n} \longrightarrow D_S^n \times \mathbb{R}^k.$$

On the overlap S^{n-1} , these two trivializations differ by a transition function

$$g : S^{n-1} \longrightarrow GL(k, \mathbb{R}).$$

Using a bundle metric and applying Gram-Schmidt, we may deform this transition function into $O(k)$. Thus we obtain a map

$$\gamma : S^{n-1} \rightarrow O(k)$$

such that

$$E \cong E_\gamma.$$

Therefore every rank k real vector bundle over S^n is obtained by a clutching construction.

It remains to discuss when two such bundles are isomorphic. Suppose first that

$$\gamma \simeq \gamma'.$$

Let

$$H : S^{n-1} \times I \longrightarrow O(k)$$

be a homotopy from γ to γ' , so that

$$H(x, 0) = \gamma(x), \quad H(x, 1) = \gamma'(x).$$

Using H , one constructs a vector bundle over

$$S^n \times I$$

whose restriction to $S^n \times \{0\}$ is E_γ , and whose restriction to $S^n \times \{1\}$ is $E_{\gamma'}$. Since I is contractible, these endpoint restrictions are isomorphic. Hence

$$E_\gamma \cong E_{\gamma'}.$$

Conversely, suppose that

$$E_\gamma \cong E_{\gamma'}.$$

An isomorphism between these bundles, when restricted to the two hemispheres, is represented by maps

$$A_N : D_N^n \rightarrow O(k), \quad A_S : D_S^n \rightarrow O(k).$$

On the equator S^{n-1} , compatibility with the clutching functions gives

$$\gamma'(x)A_N(x) = A_S(x)\gamma(x).$$

Thus

$$\gamma'(x) = A_S(x)\gamma(x)A_N(x)^{-1}.$$

Since A_N and A_S extend over disks, their restrictions to S^{n-1} are null-homotopic. Hence γ' is homotopic to γ . Therefore

$$E_\gamma \cong E_{\gamma'} \iff \gamma \simeq \gamma'.$$

Thus isomorphism classes of rank k real vector bundles over S^n are classified by homotopy classes of clutching maps

$$[S^{n-1}, O(k)].$$

7.2 Problem 12: The unique nontrivial k -plane bundle over S^1

Problem 12. Show that up to isomorphism there is a unique nontrivial k -dimensional real vector bundle E_k over S^1 . Use Problem 11 to compute

$$H^*(S(E_k)).$$

Solution.

For S^1 , the clutching construction says that rank k real vector bundles over S^1 are classified by maps into $O(k)$. Equivalently, one may describe such a bundle as a mapping torus

$$E_A = [0, 1] \times \mathbb{R}^k / (1, v) \sim (0, Av),$$

where

$$A \in O(k).$$

The orthogonal group $O(k)$ has two path components:

$$O(k) = SO(k) \sqcup \{A \in O(k) : \det A = -1\}.$$

Thus the determinant of A gives the only invariant:

$$\det A = +1 \quad \text{or} \quad \det A = -1.$$

If

$$\det A = +1,$$

then A lies in the identity component $SO(k)$, so it is homotopic to the identity. The corresponding bundle is therefore trivial:

$$E_A \cong S^1 \times \mathbb{R}^k.$$

If

$$\det A = -1,$$

then A lies in the other component of $O(k)$, and it is homotopic to a reflection, for example

$$R = \text{diag}(-1, 1, \dots, 1).$$

All such matrices give isomorphic bundles. Hence there is exactly one nontrivial rank k real vector bundle over S^1 . It is

$$E_k \cong \mu \oplus \varepsilon^{k-1},$$

where μ is the Möbius line bundle and ε^{k-1} is the trivial bundle of rank $k-1$.

Now consider the unit sphere bundle

$$S(E_k) \rightarrow S^1.$$

Since E_k is obtained from the reflection R , its sphere bundle is the mapping torus

$$S(E_k) \cong S^{k-1} \times [0, 1] / (u, 1) \sim (Ru, 0).$$

Thus $S(E_k)$ is the mapping torus of the reflection

$$R : S^{k-1} \rightarrow S^{k-1}.$$

The map R has degree

$$\deg(R) = -1.$$

Therefore the induced map on cohomology satisfies

$$R^* : H^0(S^{k-1}; \mathbb{Z}) \rightarrow H^0(S^{k-1}; \mathbb{Z})$$

is the identity, while

$$R^* : H^{k-1}(S^{k-1}; \mathbb{Z}) \rightarrow H^{k-1}(S^{k-1}; \mathbb{Z})$$

is multiplication by -1 .

Using the Wang sequence for the mapping torus, we obtain

$$\cdots \rightarrow H^{q-1}(S^{k-1}) \xrightarrow{R^* - \text{id}} H^{q-1}(S^{k-1}) \rightarrow H^q(S(E_k)) \rightarrow H^q(S^{k-1}) \xrightarrow{R^* - \text{id}} H^q(S^{k-1}) \rightarrow \cdots.$$

Because

$$H^q(S^{k-1}; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & q = 0, \\ \mathbb{Z}, & q = k-1, \\ 0, & \text{otherwise,} \end{cases}$$

the only nontrivial maps are the identity difference in degree 0, and multiplication by -2 in degree $k-1$:

$$R^* - \text{id} = -1 - 1 = -2.$$

Hence, for $k \geq 2$,

$$H^q(S(E_k); \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = 0, \\ \mathbb{Z}, & q = 1, \\ \mathbb{Z}/2, & q = k, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, additively,

$$H^*(S(E_k); \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}\alpha \oplus (\mathbb{Z}/2)\tau,$$

where

$$|\alpha| = 1, \quad |\tau| = k.$$

The class α comes from the base circle S^1 , while the class τ is the top torsion class.

As a ring, for $k \geq 2$,

$$H^*(S(E_k); \mathbb{Z}) \cong \mathbb{Z}[\alpha, \tau] / (\alpha^2, 2\tau, \alpha\tau, \tau^2),$$

with

$$|\alpha| = 1, \quad |\tau| = k.$$

For $k = 1$, the nontrivial line bundle over S^1 is the Möbius line bundle. Its sphere bundle has fiber

$$S^0 = \{-1, 1\},$$

and the reflection interchanges the two points. The total space is again a circle:

$$S(E_1) \cong S^1.$$

Therefore

$$H^*(S(E_1); \mathbb{Z}) \cong H^*(S^1; \mathbb{Z}).$$

7.3 Geometric interpretation

The nontrivial bundle E_k over S^1 is obtained by taking a cylinder

$$[0, 1] \times \mathbb{R}^k$$

and gluing the two ends with a reflection:

$$(1, v) \sim (0, Rv).$$

For $k = 1$, this gives the Möbius line bundle.

For $k = 2$, the sphere bundle $S(E_2)$ has fiber S^1 . Since the monodromy is a reflection of the circle, the total space is the Klein bottle:

$$S(E_2) \cong K.$$

Hence

$$H^0(K; \mathbb{Z}) \cong \mathbb{Z}, \quad H^1(K; \mathbb{Z}) \cong \mathbb{Z}, \quad H^2(K; \mathbb{Z}) \cong \mathbb{Z}/2.$$

For $k \geq 3$, $S(E_k)$ is a higher-dimensional analogue of the Klein bottle: it is the mapping torus of a reflection of S^{k-1} .

7.4 Summary chart

Object	Clutching data	Total space	Cohomology
Trivial bundle over S^1	$\det A = +1$	$S^1 \times \mathbb{R}^k$	trivial case
E_k	$\det A = -1$	$\mu \oplus \varepsilon^{k-1}$	unique nontrivial case
$S(E_k)$	$R : S^{k-1} \rightarrow S^{k-1}, \deg R = -1$	mapping torus of R	$H^0 = \mathbb{Z}, H^1 = \mathbb{Z}, H^k = \mathbb{Z}/2$

8 Theory background for Problems 11–12: Clutching functions and sphere bundles

8.1 Why vector bundles over spheres are controlled by equators

The basic geometric idea is that a sphere can be built from two disks:

$$S^n = D_N^n \cup D_S^n,$$

where D_N^n is the northern hemisphere and D_S^n is the southern hemisphere. Their common boundary is the equator:

$$D_N^n \cap D_S^n = S^{n-1}.$$

Since each disk is contractible, every vector bundle over each disk is trivial. Therefore, if $E \rightarrow S^n$ is a rank k real vector bundle, then

$$E|_{D_N^n} \cong D_N^n \times \mathbb{R}^k, \quad E|_{D_S^n} \cong D_S^n \times \mathbb{R}^k.$$

So the bundle has no essential twisting inside either hemisphere. All possible twisting is concentrated in the way the two trivial bundles are glued together along the equator.

This gluing is described by a transition function

$$\gamma : S^{n-1} \longrightarrow GL(k, \mathbb{R}).$$

After choosing a bundle metric, we may reduce the structure group from $GL(k, \mathbb{R})$ to $O(k)$. Thus one usually writes

$$\gamma : S^{n-1} \longrightarrow O(k).$$

The resulting bundle is called the bundle obtained from the clutching function γ .

8.2 The clutching construction

Given

$$\gamma : S^{n-1} \rightarrow O(k),$$

we form

$$E_\gamma = D_N^n \times \mathbb{R}^k \sqcup D_S^n \times \mathbb{R}^k / \sim,$$

where

$$(x, v) \sim (x, \gamma(x)v) \quad \text{for } x \in S^{n-1}.$$

The map

$$\pi : E_\gamma \rightarrow S^n$$

is defined by

$$\pi([x, v]) = x.$$

This is well-defined because the gluing identifies only points lying over the same base point x . The fiber over x is still a copy of $\mathbb{R}^k$, and the gluing maps are linear isomorphisms. Therefore E_γ is a rank k real vector bundle.

The philosophy is:

Vector bundles over S^n are obtained by gluing two trivial bundles over disks.

The nontrivial information is not inside the disks. It is in the equatorial gluing map.

8.3 Classification by homotopy classes

Two clutching maps that are homotopic give isomorphic vector bundles. Indeed, if

$$\gamma \simeq \gamma',$$

then there is a homotopy

$$H : S^{n-1} \times I \rightarrow O(k)$$

with

$$H(x, 0) = \gamma(x), \quad H(x, 1) = \gamma'(x).$$

This homotopy constructs a vector bundle over

$$S^n \times I$$

interpolating between E_γ and $E_{\gamma'}$. Since I is contractible, the two endpoint bundles are isomorphic.

Conversely, an isomorphism

$$E_\gamma \cong E_{\gamma'}$$

forces the clutching maps to differ only by changes of trivialization on the northern and southern disks. Since the disks are contractible, these changes of trivialization are null-homotopic on the equator. Hence the clutching maps represent the same homotopy class.

Thus, for $n \geq 2$, rank k real vector bundles over S^n are classified by

$$[S^{n-1}, O(k)].$$

Equivalently, choosing a base point gives

$$[S^{n-1}, O(k)] \cong \pi_{n-1}(O(k))$$

when S^{n-1} is connected.

8.4 Important special feature when $n = 1$

For S^1 , the equator is

$$S^0 = \{-1, +1\}.$$

So one must be slightly careful. A clutching map

$$S^0 \rightarrow O(k)$$

amounts to choosing two matrices in $O(k)$. But by changing trivializations over the two intervals, one may normalize one of them to the identity. What remains is a single matrix

$$A \in O(k).$$

Thus a rank k real vector bundle over S^1 can be written as a mapping torus:

$$E_A = [0, 1] \times \mathbb{R}^k / (1, v) \sim (0, Av).$$

The matrix A describes what happens to a vector when one goes once around the circle.

8.5 The two components of $O(k)$

The orthogonal group has two connected components:

$$O(k) = SO(k) \sqcup \{A \in O(k) : \det A = -1\}.$$

The determinant gives a map

$$\det : O(k) \rightarrow \{\pm 1\}.$$

Therefore there are only two possible types of rank k real vector bundles over S^1 :

$$\det A = +1 \quad \text{or} \quad \det A = -1.$$

If

$$\det A = +1,$$

then $A \in SO(k)$, and A is homotopic to the identity. Hence the bundle is trivial:

$$E_A \cong S^1 \times \mathbb{R}^k.$$

If

$$\det A = -1,$$

then A lies in the other component of $O(k)$. Every such A is homotopic to a reflection, for example

$$R = \text{diag}(-1, 1, \dots, 1).$$

Hence all such bundles are isomorphic.

So, up to isomorphism, there is exactly one nontrivial rank k real vector bundle over S^1 .

It is

$$E_k \cong \mu \oplus \varepsilon^{k-1},$$

where μ is the Möbius line bundle and ε^{k-1} is the trivial rank $k-1$ bundle.

8.6 Connection with the first Stiefel–Whitney class

The above classification can also be expressed using the first Stiefel–Whitney class. For a real vector bundle $E \rightarrow X$,

$$w_1(E) \in H^1(X; \mathbb{Z}/2)$$

detects whether E is orientable.

For $X = S^1$,

$$H^1(S^1; \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Hence there are two possibilities:

$$w_1(E) = 0$$

or

$$w_1(E) \neq 0.$$

The trivial bundle satisfies

$$w_1(S^1 \times \mathbb{R}^k) = 0.$$

The unique nontrivial bundle satisfies

$$w_1(E_k) \neq 0.$$

Thus the two classifications agree:

Rank k real vector bundles over S^1 are classified by $w_1 \in H^1(S^1; \mathbb{Z}/2)$.

8.7 The sphere bundle of a vector bundle

If $E \rightarrow B$ is a rank k real vector bundle equipped with a metric, its unit sphere bundle is

$$S(E) = \{v \in E : \|v\| = 1\}.$$

The projection

$$S(E) \rightarrow B$$

has fiber

$$S^{k-1}.$$

Thus $S(E)$ is an S^{k-1} -bundle over B .

For the unique nontrivial bundle over S^1 ,

$$E_k = \mu \oplus \varepsilon^{k-1},$$

the sphere bundle is

$$S(E_k) \cong S^{k-1} \times [0, 1] / (u, 1) \sim (Ru, 0),$$

where

$$R = \text{diag}(-1, 1, \dots, 1)$$

acts as a reflection on S^{k-1} .

Therefore $S(E_k)$ is the mapping torus of a reflection:

$$S(E_k) = T_R.$$

8.8 Mapping tori

Given a space F and a homeomorphism

$$f : F \rightarrow F,$$

the mapping torus of f is

$$T_f = F \times [0, 1] / (x, 1) \sim (f(x), 0).$$

It is a fiber bundle over S^1 :

$$F \longrightarrow T_f \longrightarrow S^1.$$

The fiber is F , and going once around the base circle applies the monodromy map f .

For $S(E_k)$, the fiber is

$$F = S^{k-1},$$

and the monodromy is the reflection

$$R : S^{k-1} \rightarrow S^{k-1}.$$

Thus the topology of $S(E_k)$ is controlled by the action of R on the cohomology of S^{k-1} .

8.9 The Wang sequence

The cohomology of a mapping torus is computed by the Wang sequence. For

$$T_f = F \times [0, 1] / (x, 1) \sim (f(x), 0),$$

there is a long exact sequence

$$\cdots \rightarrow H^{q-1}(F) \xrightarrow{f^* - \text{id}} H^{q-1}(F) \rightarrow H^q(T_f) \rightarrow H^q(F) \xrightarrow{f^* - \text{id}} H^q(F) \rightarrow \cdots.$$

In our case,

$$F = S^{k-1}.$$

The cohomology of the sphere is

$$H^q(S^{k-1}; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & q = 0, \\ \mathbb{Z}, & q = k-1, \\ 0, & \text{otherwise.} \end{cases}$$

The reflection $R : S^{k-1} \rightarrow S^{k-1}$ has degree

$$\deg(R) = -1.$$

Therefore

$$R^* = \text{id} \quad \text{on } H^0(S^{k-1}; \mathbb{Z}),$$

but

$$R^* = -\text{id} \quad \text{on } H^{k-1}(S^{k-1}; \mathbb{Z}).$$

Hence

$$R^* - \text{id} = 0$$

in degree 0, while

$$R^* - \text{id} = -\text{id} - \text{id} = -2\text{id}$$

in degree $k-1$.

This is the source of the $\mathbb{Z}/2$ -torsion in the top cohomology of $S(E_k)$.

8.10 Cohomology of $S(E_k)$

For $k \geq 2$, the Wang sequence gives

$$H^q(S(E_k); \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = 0, \\ \mathbb{Z}, & q = 1, \\ \mathbb{Z}/2, & q = k, \\ 0, & \text{otherwise.} \end{cases}$$

Thus additively,

$$H^*(S(E_k); \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}\alpha \oplus (\mathbb{Z}/2)\tau,$$

where

$$|\alpha| = 1, \quad |\tau| = k.$$

The class α comes from the base circle S^1 , while the torsion class τ comes from the orientation-reversing monodromy on the fiber sphere.

As a ring,

$$H^*(S(E_k); \mathbb{Z}) \cong \mathbb{Z}[\alpha, \tau] / (\alpha^2, 2\tau, \alpha\tau, \tau^2),$$

with

$$|\alpha| = 1, \quad |\tau| = k.$$

8.11 Low-dimensional geometric cases

For $k = 1$, the nontrivial bundle is the Möbius line bundle

$$\mu \rightarrow S^1.$$

Its sphere bundle has fiber

$$S^0 = \{-1, +1\}.$$

Because the Möbius twist exchanges the two points of the fiber, the total space is a single circle:

$$S(\mu) \cong S^1.$$

For $k = 2$,

$$E_2 \cong \mu \oplus \varepsilon^1.$$

The sphere bundle has fiber

$$S^1.$$

The monodromy is a reflection of the circle. Therefore

$$S(E_2)$$

is the Klein bottle. This is a central example: the Klein bottle is an S^1 -bundle over S^1 with orientation-reversing monodromy.

For $k \geq 3$, $S(E_k)$ is a higher-dimensional analogue of the Klein bottle. It is an S^{k-1} -bundle over S^1 whose monodromy reverses the orientation of the fiber.

8.12 Summary

The central ideas are:

Vector bundles over S^n are classified by clutching maps $S^{n-1} \rightarrow O(k)$.

Over S^1 , rank k real vector bundles have only two types: trivial and nontrivial.

The unique nontrivial bundle is $E_k \cong \mu \oplus \varepsilon^{k-1}$.

$S(E_k)$ is the mapping torus of a reflection of S^{k-1} .

$$H^q(S(E_k); \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = 0, \\ \mathbb{Z}, & q = 1, \\ \mathbb{Z}/2, & q = k, \\ 0, & \text{otherwise,} \end{cases} \quad k \geq 2.$$

9 Concrete examples for Problems 11–12

9.1 Example 1: The trivial bundle over S^n

Take the clutching function

$$\gamma : S^{n-1} \rightarrow O(k)$$

to be constant:

$$\gamma(x) = I_k$$

for every $x \in S^{n-1}$.

Then the gluing relation is

$$(x, v) \sim (x, I_k v) = (x, v).$$

Thus no twisting occurs along the equator. Therefore

$$E_\gamma \cong S^n \times \mathbb{R}^k.$$

This is the trivial rank k vector bundle over S^n .

Geometrically, we glue the two product bundles

$$D_N^n \times \mathbb{R}^k$$

and

$$D_S^n \times \mathbb{R}^k$$

without rotating or reflecting the fibers.

Hence

$$\gamma \simeq I_k \implies E_\gamma \text{ is trivial.}$$

9.2 Example 2: The Möbius line bundle over S^1

Let

$$k = 1.$$

Then

$$O(1) = \{+1, -1\}.$$

A rank 1 real vector bundle over S^1 can be written as

$$E_a = [0, 1] \times \mathbb{R} / (1, v) \sim (0, av),$$

where

$$a \in O(1).$$

There are two possibilities.

If

$$a = +1,$$

then

$$(1, v) \sim (0, v),$$

and we obtain the cylinder:

$$E_{+1} \cong S^1 \times \mathbb{R}.$$

This is the trivial line bundle.

If

$$a = -1,$$

then

$$(1, v) \sim (0, -v).$$

This reverses the fiber direction after going once around the base circle. The resulting bundle is the Möbius line bundle:

$$E_{-1} \cong \mu.$$

Thus

Real line bundles over S^1 are the cylinder and the Möbius band.

9.3 Example 3: The unique nontrivial rank 2 bundle over S^1

Let

$$k = 2.$$

The nontrivial bundle is obtained from the reflection

$$R = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$E_2 = [0, 1] \times \mathbb{R}^2 / (1, v) \sim (0, Rv).$$

Writing

$$v = (x, y),$$

we have

$$R(x, y) = (-x, y).$$

Thus, after going once around the base circle, the first coordinate changes sign, while the second coordinate does not. Therefore

$$E_2 \cong \mu \oplus \varepsilon^1,$$

where μ is the Möbius line bundle and ε^1 is the trivial line bundle.

Hence

$$E_2 = \text{one Möbius direction plus one trivial direction.}$$

This is the unique nontrivial rank 2 real vector bundle over S^1 .

9.4 Example 4: Sphere bundle of the trivial rank 2 bundle over S^1

Consider the trivial rank 2 bundle

$$\varepsilon^2 = S^1 \times \mathbb{R}^2.$$

Its unit sphere bundle is

$$S(\varepsilon^2) = S^1 \times S^1.$$

Indeed, over every point of the base S^1 , the fiber is $\mathbb{R}^2$, and the unit sphere in $\mathbb{R}^2$ is

$$S^1.$$

Therefore the total space is

$$S^1 \times S^1 = T^2.$$

Hence

$$S(\varepsilon^2) \cong T^2.$$

The cohomology of the torus is

$$H^0(T^2; \mathbb{Z}) \cong \mathbb{Z},$$

$$H^1(T^2; \mathbb{Z}) \cong \mathbb{Z}^2,$$

$$H^2(T^2; \mathbb{Z}) \cong \mathbb{Z}.$$

Thus the trivial rank 2 bundle over S^1 gives the torus as its sphere bundle.

9.5 Example 5: Sphere bundle of the nontrivial rank 2 bundle over S^1

Now take

$$E_2 = \mu \oplus \varepsilon^1.$$

Its sphere bundle has fiber

$$S^1.$$

The monodromy is the reflection

$$R : S^1 \rightarrow S^1.$$

Therefore

$$S(E_2) = S^1 \times [0, 1] / (u, 1) \sim (Ru, 0).$$

This is the mapping torus of a reflection of S^1 . Geometrically, this total space is the Klein bottle:

$$\boxed{S(E_2) \cong K.}$$

So we have the important comparison

$$S(\varepsilon^2) \cong T^2,$$

but

$$S(E_2) \cong K.$$

The trivial bundle gives a torus, while the nontrivial bundle gives a Klein bottle.

Their integral cohomology groups are different:

$$H^*(T^2; \mathbb{Z}) : \quad H^0 = \mathbb{Z}, \quad H^1 = \mathbb{Z}^2, \quad H^2 = \mathbb{Z}.$$

For the Klein bottle,

$$H^*(K; \mathbb{Z}) : \quad H^0 = \mathbb{Z}, \quad H^1 = \mathbb{Z}, \quad H^2 = \mathbb{Z}/2.$$

Thus the reflection in the clutching function changes the topology of the sphere bundle dramatically.

9.6 Example 6: Sphere bundle of the nontrivial rank 3 bundle over S^1

Let

$$E_3 = \mu \oplus \varepsilon^2.$$

The clutching matrix is

$$R = \text{diag}(-1, 1, 1).$$

Hence

$$E_3 = [0, 1] \times \mathbb{R}^3 / (1, v) \sim (0, Rv).$$

The sphere bundle has fiber

$$S^2.$$

Therefore

$$S(E_3) = S^2 \times [0, 1] / (u, 1) \sim (Ru, 0).$$

This is the mapping torus of a reflection of S^2 .

The reflection reverses orientation on S^2 , so the induced map on top cohomology is multiplication by -1 :

$$R^* : H^2(S^2; \mathbb{Z}) \rightarrow H^2(S^2; \mathbb{Z}), \quad R^* = -1.$$

The Wang sequence gives

$$\begin{aligned} H^0(S(E_3); \mathbb{Z}) &\cong \mathbb{Z}, \\ H^1(S(E_3); \mathbb{Z}) &\cong \mathbb{Z}, \\ H^3(S(E_3); \mathbb{Z}) &\cong \mathbb{Z}/2, \end{aligned}$$

and all other cohomology groups vanish.

Therefore

$$H^q(S(E_3); \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = 0, \\ \mathbb{Z}, & q = 1, \\ \mathbb{Z}/2, & q = 3, \\ 0, & \text{otherwise.} \end{cases}$$

This is a 3-dimensional analogue of the Klein bottle.

9.7 Example 7: Real line bundles over S^n for $n \geq 2$

A real line bundle has rank

$$k = 1.$$

So its clutching function is

$$\gamma : S^{n-1} \rightarrow O(1).$$

But

$$O(1) = \{+1, -1\}.$$

For $n \geq 2$, the equator

$$S^{n-1}$$

is connected. A continuous map from a connected space to the discrete space $\{+1, -1\}$ must be constant.

Therefore every clutching map

$$\gamma : S^{n-1} \rightarrow O(1)$$

is constant.

Hence every real line bundle over S^n , for $n \geq 2$, is trivial:

Every real line bundle over S^n is trivial for $n \geq 2$.

For example,

every real line bundle over S^2 is trivial,

and

every real line bundle over S^3 is trivial.

This contrasts with S^1 , where the nontrivial Möbius line bundle exists.

9.8 Example 8: Oriented rank 2 bundles over S^2

Now consider oriented rank 2 real vector bundles over S^2 .

The structure group is

$$SO(2).$$

Since

$$SO(2) \cong S^1,$$

the clutching construction says that oriented rank 2 bundles over S^2 are classified by

$$[S^1, SO(2)].$$

Using

$$SO(2) \cong S^1,$$

we obtain

$$[S^1, S^1] \cong \mathbb{Z}.$$

The integer is the winding number of the clutching function.

A typical clutching function is

$$\gamma_m : S^1 \rightarrow SO(2),$$

defined by

$$\gamma_m(e^{i\theta}) = \begin{pmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{pmatrix}.$$

The integer

$$m \in \mathbb{Z}$$

measures how many times the fiber rotates while going once around the equator.

Thus

Oriented rank 2 real bundles over S^2 are classified by an integer.

That integer is the Euler number of the bundle.

9.9 Example 9: The tangent bundle of S^2

The tangent bundle

$$TS^2 \rightarrow S^2$$

is a rank 2 oriented real vector bundle.

Using the clutching construction, write

$$S^2 = D_N^2 \cup D_S^2,$$

where the overlap is the equator

$$S^1.$$

The tangent bundle is trivial over each hemisphere. Therefore it is determined by a transition function

$$\gamma : S^1 \rightarrow SO(2).$$

For TS^2 , this clutching function has degree 2. In other words, as one goes once around the equator, the tangent frame rotates twice.

Thus

$$TS^2$$

corresponds to the integer

$$2 \in \mathbb{Z}.$$

This agrees with the Euler characteristic:

$$e(TS^2) = \chi(S^2) = 2.$$

Therefore

TS^2 is nontrivial because its clutching function has degree 2.

This also explains geometrically why there is no nowhere-zero continuous tangent vector field on S^2 . If TS^2 were trivial, its clutching degree would be 0, not 2.

9.10 Example 10: The tangent bundle of S^1

The tangent bundle of the circle is

$$TS^1 \rightarrow S^1.$$

Since S^1 has a global nonzero tangent vector field, namely the unit tangent vector moving counterclockwise, we have

$$TS^1 \cong S^1 \times \mathbb{R}.$$

Therefore

$$TS^1$$

is trivial.

This agrees with the classification of line bundles over S^1 : the tangent bundle TS^1 is orientable, hence it is the trivial line bundle, not the Möbius bundle.

Thus

$$\boxed{TS^1 \cong \varepsilon^1.}$$

9.11 Example 11: Comparing TS^1 , TS^2 , and TS^3

The tangent bundle of S^1 is trivial:

$$TS^1 \cong S^1 \times \mathbb{R}.$$

The tangent bundle of S^2 is nontrivial:

$$TS^2 \not\cong S^2 \times \mathbb{R}^2.$$

The tangent bundle of S^3 is trivial:

$$TS^3 \cong S^3 \times \mathbb{R}^3.$$

The reason is that S^3 is a Lie group:

$$S^3 \cong SU(2),$$

and every Lie group has a trivial tangent bundle.

Thus the pattern is not simply that all spheres have nontrivial tangent bundle. Instead,

S^1 is parallelizable,

S^2 is not parallelizable,

S^3 is parallelizable.

In fact, the only spheres with trivial tangent bundle are

$$S^0, \quad S^1, \quad S^3, \quad S^7.$$

9.12 Example 12: A clutching map of degree m over S^2

Let

$$\gamma_m : S^1 \rightarrow SO(2)$$

be defined by

$$\gamma_m(e^{i\theta}) = \begin{pmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{pmatrix}.$$

Then

$$E_{\gamma_m} \rightarrow S^2$$

is an oriented rank 2 real vector bundle.

The integer m is the winding number of γ_m .

Some important cases are:

$$m = 0 : \quad E_{\gamma_0} \cong S^2 \times \mathbb{R}^2.$$

$$m = 1 : \quad E_{\gamma_1} \text{ is the underlying real bundle of the Hopf line bundle.}$$

$$m = 2 : \quad E_{\gamma_2} \cong TS^2.$$

Thus

$\gamma_m \text{ gives a rank 2 bundle over } S^2 \text{ with Euler number } m.$

9.13 Example 13: Why homotopic clutching maps give the same bundle

Let

$$\gamma_0, \gamma_1 : S^{n-1} \rightarrow O(k)$$

be clutching functions.

Suppose there is a homotopy

$$H : S^{n-1} \times [0, 1] \rightarrow O(k)$$

with

$$H(x, 0) = \gamma_0(x), \quad H(x, 1) = \gamma_1(x).$$

Then, while t moves from 0 to 1, the gluing rule changes continuously.

At $t = 0$, the gluing rule gives

$$E_{\gamma_0}.$$

At $t = 1$, the gluing rule gives

$$E_{\gamma_1}.$$

Since the gluing rule changes continuously, the isomorphism type of the vector bundle does not change. Therefore

$$E_{\gamma_0} \cong E_{\gamma_1}.$$

For example, if

$$A \in SO(k),$$

then A is path-connected to the identity matrix. Hence the mapping-torus bundle

$$[0, 1] \times \mathbb{R}^k / (1, v) \sim (0, Av)$$

is trivial.

But if

$$\det A = -1,$$

then A cannot be connected to the identity inside $O(k)$. Therefore the corresponding bundle over S^1 is nontrivial.

9.14 Example 14: Torus versus Klein bottle from sphere bundles

This is the most important geometric comparison in the case of rank 2 bundles over S^1 .

The trivial case

If the monodromy is

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

then

$$E_I = S^1 \times \mathbb{R}^2.$$

The sphere bundle is

$$S(E_I) = S^1 \times S^1 = T^2.$$

Thus no reflection in the fiber gives the torus.

The nontrivial case

If the monodromy is

$$R = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

then

$$E_R = \mu \oplus \varepsilon^1.$$

The sphere bundle is

$$S(E_R) = S^1 \times [0, 1] / (u, 1) \sim (Ru, 0).$$

This is the Klein bottle:

$$S(E_R) \cong K.$$

Therefore

No twist in the sphere fiber gives T^2 .

Reflection twist in the sphere fiber gives the Klein bottle.

This is exactly the same idea as the difference between a cylinder and a Möbius band, but one dimension higher.

9.15 Summary table

Base	Rank	Clutching data	Bundle / sphere bundle
S^1	1	+1	trivial line bundle
S^1	1	-1	Möbius line bundle
S^1	2	I	$S(E) = T^2$
S^1	2	R	$S(E) = \text{Klein bottle}$
S^1	k	$\det A = +1$	trivial rank k bundle
S^1	k	$\det A = -1$	$E_k = \mu \oplus \varepsilon^{k-1}$
S^2	1	$S^1 \rightarrow O(1)$	only trivial line bundle
S^2	2	$S^1 \rightarrow SO(2)$ of degree m	Euler number m
S^2	2	degree 2	TS^2

9.16 Main examples to remember

$$E_1 = \mu.$$

$$E_k = \mu \oplus \varepsilon^{k-1}.$$

$$S(E_2) = \text{Klein bottle.}$$

$$TS^2 \text{ has clutching degree 2.}$$

10 Diagrams and summary charts for Problems 11–12

The following diagrams summarize the geometric content of the clutching construction, the unique nontrivial bundle over S^1 , and the associated sphere bundles. They are intended to support the solutions of Problems 11–12 by showing how the abstract gluing data is represented geometrically.

Vector bundles over spheres: key geometric diagrams

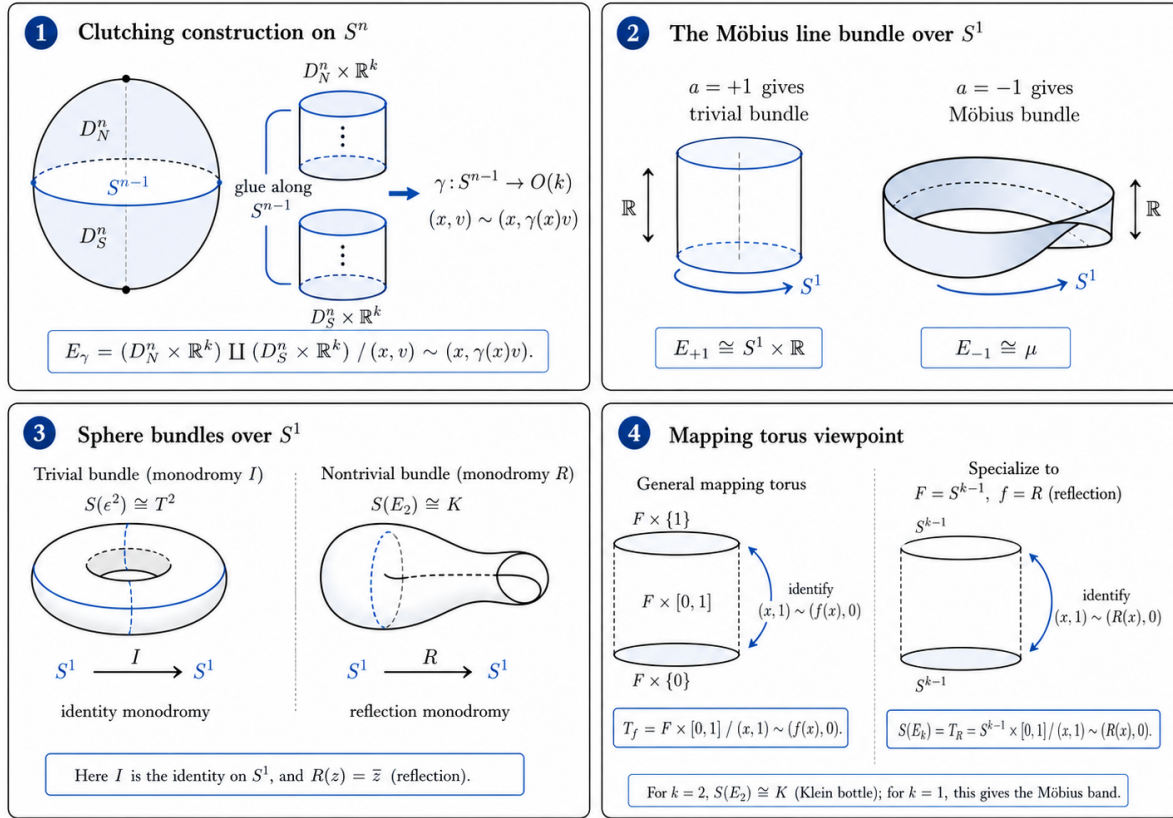
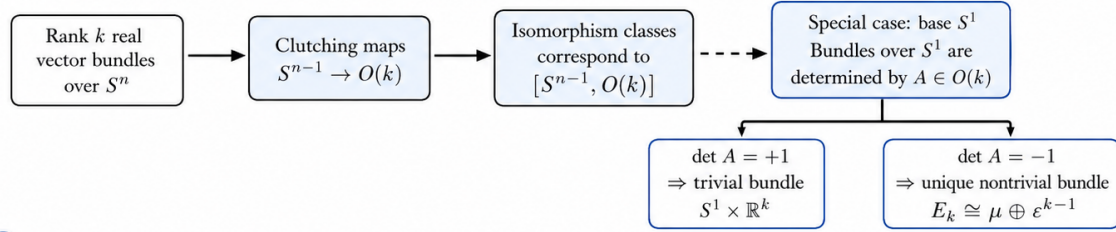


Figure 1: Key geometric diagrams for vector bundles over spheres. The first panel shows the clutching construction over S^n ; the second compares the trivial line bundle over S^1 with the Möbius line bundle; the third compares the torus and Klein bottle as sphere bundles over S^1 ; the fourth interprets these examples as mapping tori.

Classification and examples: vector bundles over spheres

1 Classification flow chart



2 Examples and comparison

Base	Rank	Clutching data	Result
S^1	1	$+1 \in O(1)$	trivial line bundle $S^1 \times \mathbb{R}$
S^1	1	$-1 \in O(1)$	Möbius line bundle μ
S^1	2	$I \in O(2)$ ($\det I = +1$)	sphere bundle = torus $T^2 = S^1 \times S^1$ (trivial)
S^1	2	$R = \text{diag}(1, -1) \in O(2)$ ($\det R = -1$)	sphere bundle = Klein bottle K
S^2	1	$S^1 \rightarrow O(1) = \{\pm 1\}$	only trivial line bundle
S^2	2	degree m map $S^1 \rightarrow SO(2) \cong S^1$	disk (or sphere) bundle with Euler number m (total space = $D(E_m)$; sphere bundle $S(E_m)$)
S^2	2	degree 2 map $S^1 \rightarrow SO(2)$	TS^2 (tangent bundle of S^2)

3 Cohomology of $S(E_k)$ for $k \geq 2$

For the unique nontrivial rank- k bundle E_k over S^1 ($E_k \cong \mu \oplus \varepsilon^{k-1}$), the cohomology of its unit sphere bundle $S(E_k)$ is	$H^q(S(E_k); \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & q = 0 \\ \mathbb{Z} & q = 1 \\ \mathbb{Z}/2 & q = k \\ 0 & \text{otherwise.} \end{cases}$	Examples: $S(E_2) \cong$ Klein bottle K. $S(E_3) =$ mapping torus of a reflection of S^2.
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Figure 2: Classification chart and examples for vector bundles over spheres. Rank k real vector bundles over S^n are described by clutching maps $S^{n-1} \rightarrow O(k)$. The special case of bundles over S^1 reduces to the determinant of the monodromy matrix $A \in O(k)$. The chart also records the cohomology of the unit sphere bundle $S(E_k)$ for the unique nontrivial bundle $E_k \rightarrow S^1$.

How to read the diagrams

In Figure 1, the bundle $E_\gamma \rightarrow S^n$ is obtained by gluing two trivial bundles over the northern and southern hemispheres. The gluing is performed along the equator S^{n-1} , and the gluing rule is encoded by the clutching map

$$\gamma : S^{n-1} \longrightarrow O(k).$$

Thus the whole bundle is determined by how the fibers are identified along the equator.

For bundles over S^1 , the monodromy matrix $A \in O(k)$ determines the bundle. If $\det A = +1$, the bundle is trivial. If $\det A = -1$, one obtains the unique nontrivial rank k bundle

$$E_k \cong \mu \oplus \varepsilon^{k-1}.$$

Its unit sphere bundle is the mapping torus of a reflection of S^{k-1} :

$$S(E_k) \cong S^{k-1} \times [0, 1] / (u, 1) \sim (Ru, 0).$$

For $k = 2$, this gives the Klein bottle:

$$S(E_2) \cong K.$$